

Total No. of Questions : 10]

SEAT No. :

P-9342

[Total No. of Pages : 2

[6181]-577

B.E. (Mechanical)

AUTOMOBILE ENGINEERING

(2015 Pattern) (Semester - I) (402045A) (Elective - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8 and Q.9 or Q.10.
- 2) Neat diagrams must draw wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Write about vehicle specifications. **[5]**

b) Describe the necessity of gearbox in automobile. **[5]**

OR

Q2) a) Discuss various types of frames used in automotive construction. **[5]**

b) Explain the working principle and applications of a torque converter. **[5]**

Q3) a) Explain construction of wheel with net sketch. **[5]**

b) Describe the fundamental principles of a vehicle's steering mechanism. **[5]**

OR

Q4) a) Explain collapsible steering. **[5]**

b) What are the functions and objectives of suspension system? **[5]**

Q5) a) Explain the factors influencing the stability of vehicles. **[8]**

b) Discuss the impact of NVH on vehicle comfort and safety. **[8]**

OR

Q6) a) Define and explain the key parameters used to evaluate vehicle performance. **[8]**

b) Describe the role of active safety and passive safety in automobile with suitable example. **[8]**

P.T.O.

- Q7)** a) Describe various test on battery conditions. [8]
b) Identify and discuss the various resistances that a vehicle encounters during motion. [8]

OR

- Q8)** a) Discuss the importance of maintaining the suspension system. [8]
b) Explain the maintenance procedures for a vehicle's brake system. [8]

- Q9)** a) How can the transition to electric and hybrid vehicles benefit or challenge various sectors of the economy? [6]
b) Explore the factors influencing consumer adoption of EVs and HEVs. [6]
c) How do these vehicles contribute to reducing carbon emissions and environmental. [6]

OR

- Q10)** a) How do these vehicles contribute to reducing carbon emissions and environmental conservation? [6]
b) Discuss the importance of developing charging infrastructure for EVs and PHEVs. [6]
c) Explain the layout, construction, and working principles of electric vehicles. [6]

